

2.4

SECTION TITLE

Electron Stability & Valence Electrons

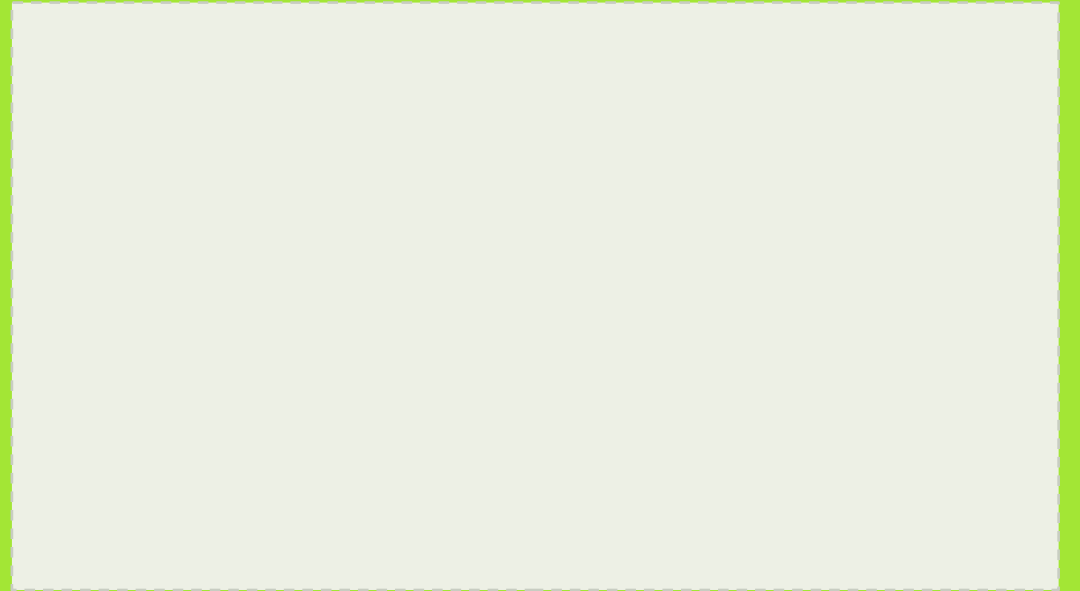
Count valence electrons,
explain why noble gases are
stable, write ion configurations,
and predict main-group ion
charges.

Who Reacts, Who Doesn't

Helium, neon, and argon barely react with anything.
Sodium and chlorine react easily.

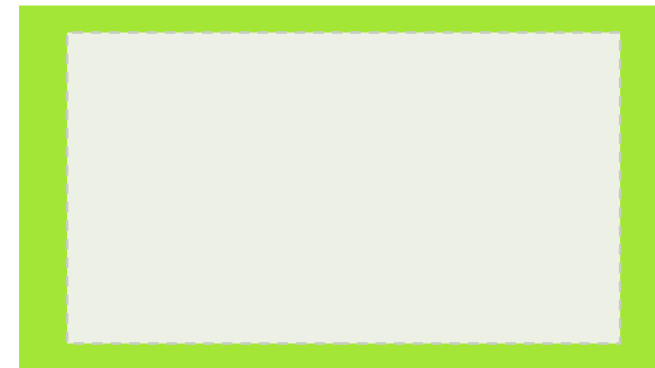
KEY QUESTION

What is different about the electrons in He, Ne, and Ar?



Three Words You Need First

Write these before we go further. Everything after this slide uses them.



VALENCE ELECTRON

An electron in the outermost energy level.

CORE ELECTRON

An electron in an inner, filled energy level.

NOBLE GAS CONFIGURATION

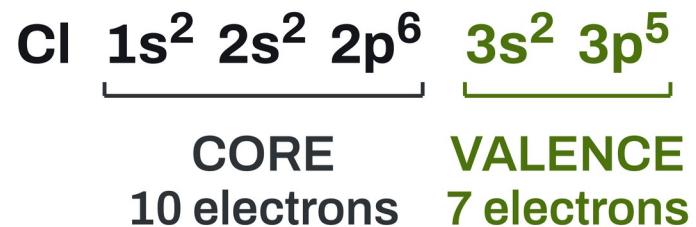
Outer s and p sublevels full: 2 or 8 electrons.

MUST WRITE

Valence electrons are the electrons in the outermost energy level, the one with the highest n.

Core or Valence? Find the Highest n

Chlorine has 17 electrons. Only the ones in $n = 3$ are valence.



$$10 + 7 = 17$$

CORE

$1s^2 2s^2 2p^6$ fill $n = 1$ and $n = 2$. That is 10 electrons.

VALENCE

$3s^2 3p^5$ is the highest level, $n = 3$. That is 7 electrons.

THE CHECK

$10 + 7 = 17$, the atomic number of chlorine.

Valence Electrons From the Table

For main-group elements, the group number gives you the count.

Group	1	2	13	14	15	16	17	18
Atom	Na	Mg	Al	Si	P	S	Cl	Ar
Valence	1	2	3	4	5	6	7	8

Helium: group 18, but 2 valence electrons
Period 3 shown. Groups 3 to 12 skipped.

THE PATTERN

Groups 1 and 2: the group number.

Groups 13 to 18: the units digit.

THE EXCEPTION

Helium is in group 18 and has 2 valence electrons, not 8.

WHAT THIS SKIPS

Groups 3 to 12, the transition metals. Their ions follow the rule on slide 13.

Example: Valence Electrons of P and Ca

We do this one together. Copy the setup, not just the answer.

PROBLEM

Use each configuration to find the number of valence electrons in phosphorus and calcium.

GIVEN

P: [Ne] 3s² 3p³

Ca: [Ar] 4s²

NEED

Highest n for each

Valence electron count

Check against the group

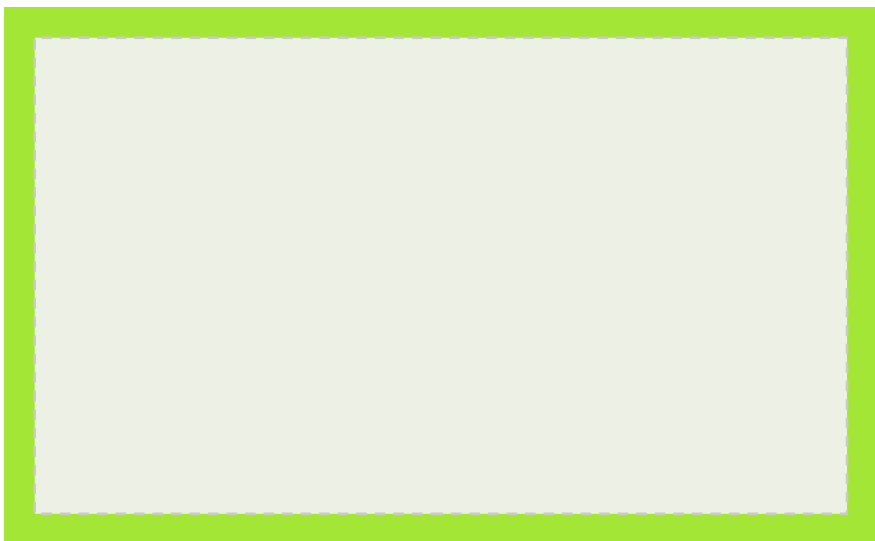
WORK AREA

MUST WRITE

Valence electrons are the s and p electrons in the highest n only.

Phosphorus and Calcium, Worked

Check the group number last. It should agree with your count.



SETUP

Find the highest n .
Count every electron in it.

WORK

P: $n = 3$, so $3s^2 3p^3$

$2 + 3 = 5$ valence

Ca: $n = 4$, so $4s^2$

2 valence

MUST WRITE

P has 5 valence electrons (group 15). Ca has 2 (group 2).

Why Noble Gases Barely React

Look at the outer s and p sublevels of each one.

	Outer s, p
He $1s^2$	2
Ne $[\text{He}] 2s^2 2p^6$	8
Ar $[\text{Ne}] 3s^2 3p^6$	8
Kr $[\text{Ar}] 4s^2 3d^{10} 4p^6$	8

OCTET AND DUET

Ne, Ar, and Kr have 8 electrons in their outer s and p sublevels. Helium has 2.

WHAT FULL MEANS

Full means s and p. Argon's 3d sublevel is empty, so $n = 3$ is not completely full.

HOW WE USE IT

Later in this section we compare ions to these four configurations.

What Makes an Ion

Atoms become ions by gaining or losing electrons. The nucleus does not change.

CATION

Loses electrons.

Positive charge.

$\text{Na} \rightarrow \text{Na}^+$: 11 electrons become 10.

Charge: 1+

ANION

Gains electrons.

Negative charge.

$\text{Cl} \rightarrow \text{Cl}^-$: 17 electrons become 18.

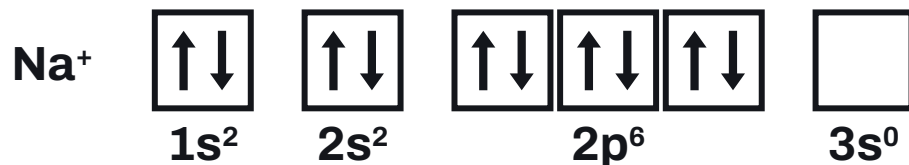
Charge: 1-

MUST WRITE

Electrons = atomic number – charge. Protons never change.

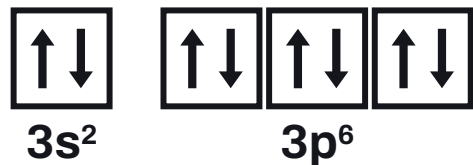
Na Loses One, Cl Gains One

Electrons leave from the highest n first. A gained electron fills the open 3p spot.



Na⁺: charge 1+, 10 electrons

**Cl⁻
[Ne]**



Cl ion: charge 1-, 18 electrons

SODIUM

3s¹ leaves: 11 → 10 electrons.

Na⁺: charge 1+. Matches [Ne].

CHLORINE

One electron fills 3p: 17 → 18.

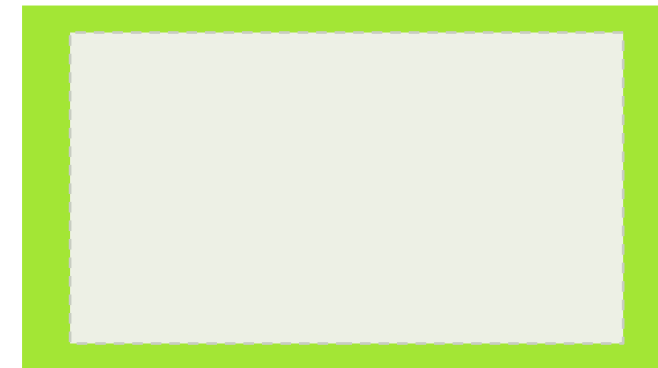
Cl⁻: charge 1-. Matches [Ar].

THE CHECK

11 - 1 = 10 and 17 + 1 = 18.

Three More Ions, Same Move

Remove from the highest n, or fill the open spots in s and p.



ALUMINUM

Al: [Ne] $3s^2 3p^1$

Al³⁺: [Ne]

13 → 10 electrons

Charge: 3+

OXYGEN

O: $1s^2 2s^2 2p^4$

O²⁻: $1s^2 2s^2 2p^6$

= [Ne], 10 electrons

Charge: 2-

BROMINE

Br: [Ar] $4s^2 3d^{10} 4p^5$

Br⁻: [Ar] $4s^2 3d^{10} 4p^6$

= [Kr], 36 electrons

Charge: 1-

MUST WRITE

A cation loses from the highest n. An anion fills open spots in s and p.

Same Electrons, Different Atoms

Isoelectronic means the same number of electrons and the same configuration.

O²⁻	F⁻	Ne	Na⁺	Mg²⁺	Al³⁺
8 p	9 p	10 p	11 p	12 p	13 p
10 e	10 e	10 e	10 e	10 e	10 e

p = protons e = electrons

ALL TEN

Each of these has 10 electrons: $1s^2 2s^2 2p^6$.

DIFFERENT NUCLEI

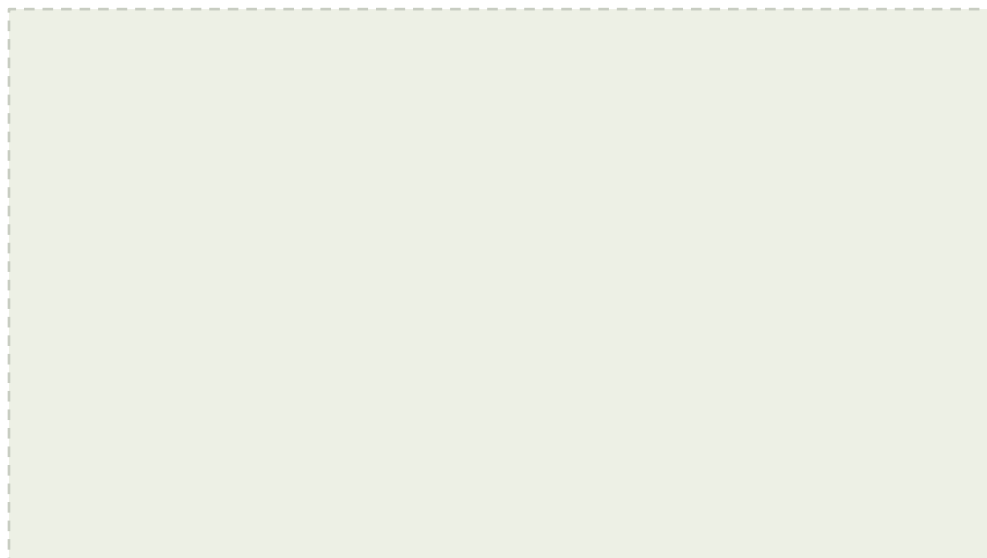
The proton counts differ, so these are different particles.
Their sizes differ too. Unit 3 explains why.

COUNT IT

Electrons = protons – charge. Check any column.

Transition Metal Ions: 4s Leaves First

This is the rule that surprises people. Watch the order.



THE RULE

Remove electrons from the outermost s first: 4s.

Then take from 3d if more must go.

THE REASON

Once 3d has electrons, it sits lower in energy than 4s.

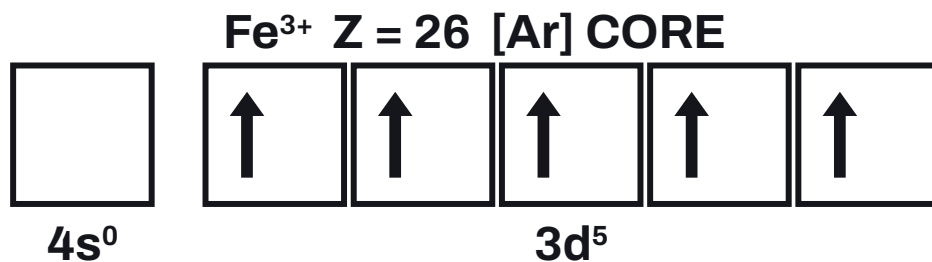
So 4s electrons leave first, even though 4s filled first.

MUST WRITE

Transition-metal ions lose their ns electrons before their (n–1)d electrons.

Iron: Fe, Fe²⁺, and Fe³⁺

Both 4s electrons go first. The third comes out of 3d.



Highest n is 4: 4s leaves first

Fe: charge 0, 26 electrons

Fe³⁺: charge 3+, 23 electrons

Fe

[Ar] 4s² 3d⁶

26 electrons

Fe²⁺

[Ar] 3d⁶

24 electrons. Both 4s electrons left.

Fe³⁺

[Ar] 3d⁵

23 electrons. One 3d electron leaves too.

Example: Zn^{2+} , Cu^+ , and Cu^{2+}

Same rule as iron. Copy the setup, not just the answer.

PROBLEM

Write the [Ar] shorthand for Zn^{2+} , Cu^+ , and Cu^{2+} .

GIVEN

Zn: $[\text{Ar}] 4s^2 3d^{10}$

Cu: $[\text{Ar}] 4s^1 3d^{10}$

NEED

Electrons to remove

Which sublevel they
leave

Each ion's shorthand

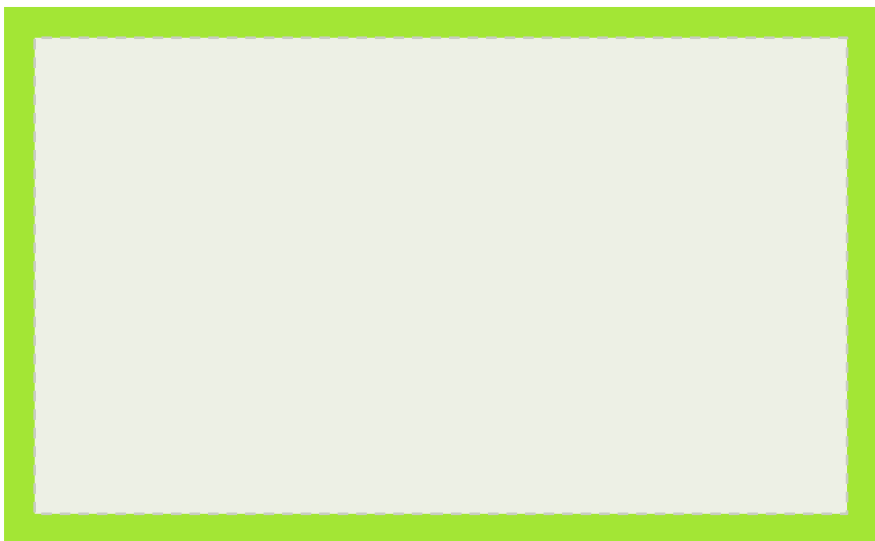
WORK AREA

MUST WRITE

Take the 4s electrons first. Only then take from 3d.

Zn²⁺, Cu⁺, and Cu²⁺, Worked

None of these ends at a noble gas configuration.



SETUP

Take 4s first, then 3d.

Zn 30 → 28. Cu 29 → 28 or 27.

WORK

Zn²⁺: [Ar] 3d¹⁰

Cu⁺: [Ar] 3d¹⁰

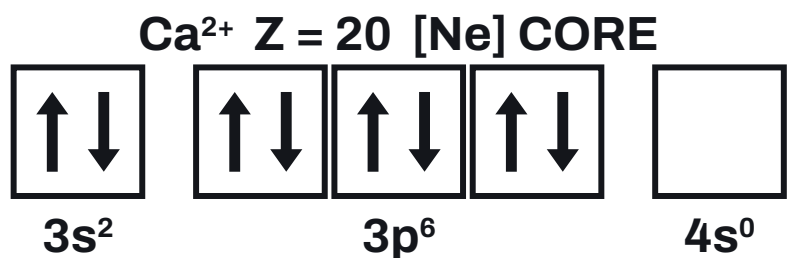
Cu²⁺: 4s, then one 3d
→ [Ar] 3d⁹

MUST WRITE

Zn²⁺ and Cu⁺ are both [Ar] 3d¹⁰. Cu²⁺ is [Ar] 3d⁹.

Why Calcium Forms Ca^{2+}

Count valence electrons first. The count tells you the move.



Valence electrons leave first: this is the outermost level

Ca: charge 0, 20 electrons

$\text{Ca}^{2+} = [\text{Ar}]$: charge 2+

THE ATOM

Ca: $[\text{Ar}] 4s^2$

2 valence electrons.

Lose 2, not gain 6: the fewest wins.

THE ION

Lose both 4s electrons: $[\text{Ar}]$.

18 electrons, 20 protons.

Charge: $20 - 18 = 2+$

MUST WRITE

Ca has 2 valence electrons. Losing both leaves $[\text{Ar}]$, so Ca forms Ca^{2+} .

Valence Electrons Predict the Charge

Lose or gain the fewest electrons to reach 8. Unit 3 will explain the energy side.

Group	Atom	Valence	Move	Charge
1	Na	1	lose 1	1+
2	Mg	2	lose 2	2+
13	Al	3	lose 3	3+
14	Si	4	not predicted	—
15	P	5	gain 3	3-
16	S	6	gain 2	2-
17	Cl	7	gain 1	1-
18	Ar	8	none	0

LOSE

Groups 1, 2, 13 lose all their valence electrons.
Charges 1+, 2+, 3+.

GAIN

Groups 15, 16, 17 gain 3, 2, 1 to reach 8.
Charges 3-, 2-, 1-.

NOT PREDICTED HERE

Group 14 and the transition metals. Group 18 is already full.

Example: Predict the Ions of Mg, Al, S, Br

We do this one together. Copy the setup, not just the answer.



PROBLEM

Use valence electrons to predict each ion. Give the move, the result, and the charge.

GIVEN

Mg: group 2
Al: group 13
S: group 16
Br: group 17

NEED

Valence electrons
Lose or gain how many
Noble gas and charge

WORK AREA

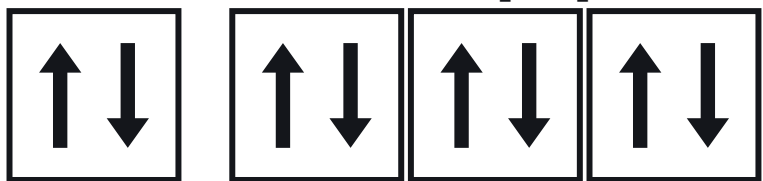
MUST WRITE

Lose or gain the fewest electrons to reach 8 valence electrons.

Mg, Al, S, and Br Ions, Worked

Count, move, result, charge. Check the electron count at the end.

SULFUR ION $Z = 16$ [Ne] CORE

 $3s^2$ $3p^6$

6 valence + 2 gained = 8

S ion: charge 2-

SETUP

Each line reads: valence,
move, noble gas, charge.

WORK

Mg: 2, lose 2, [Ne], 2+

Al: 3, lose 3, [Ne], 3+

S: 6, gain 2, [Ar], 2-

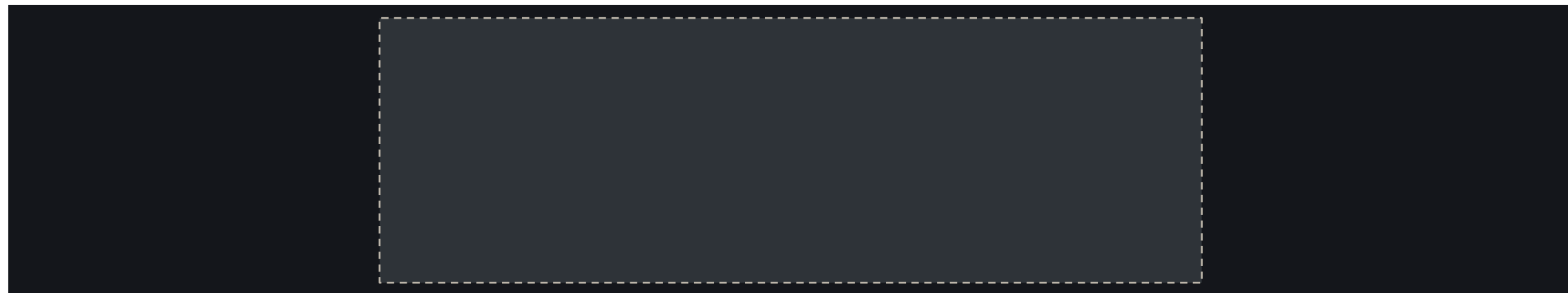
Br: 7, gain 1, [Kr], 1-

MUST WRITE

Mg^{2+} , Al^{3+} , S^{2-} , Br^- . Each move ends at a noble gas configuration.

Check Yourself

Work these in your notes. We compare in a minute.



1 Valence

Valence electrons in Al, Se, and Xe?

2 Ca^{2+}

Write it. Name the matching noble gas.

3 Predict

Predict O's ion. Write its full configuration.

4 Three ions

[Ar] shorthand for Fe^{3+} , Cu^{2+} , and Zn^{2+} .

LOOKING AHEAD

Unit 3 Picks Up From Here

You can predict the charge of a main-group ion from its valence electrons. Unit 3 explains the energy side.